MCU System Clock & RCC

Embedded Controller

Prof. Young-Keun Kim

MCU System Clock

- **Contents**
  - What is system clock
  - External and Internal Clock
  - PLL
  - Real Time Clock
  - Clock for Peripheral
  - Reset & Clock Control (RCC) Register

MCU System Clock

- **System Clocks (SYSCLK)**
  - 3 different clock sources to drive system clocks
    - ✓ HSI oscillator clock : RC based, Internal Oscillator in CPU (default)
    - ✓ HSE oscillator clock : External Crystal (OFF at reset)
    - ✓ Main PLL clock

- **Secondary clock sources**
  - LSI(Low-Speed Internal): RC based, internal MCU, 32kHz
  - LSE(Low-Speed External): 32.768kHz low-speed external crystal to drive Real-Time-Clock

MCU System Clock

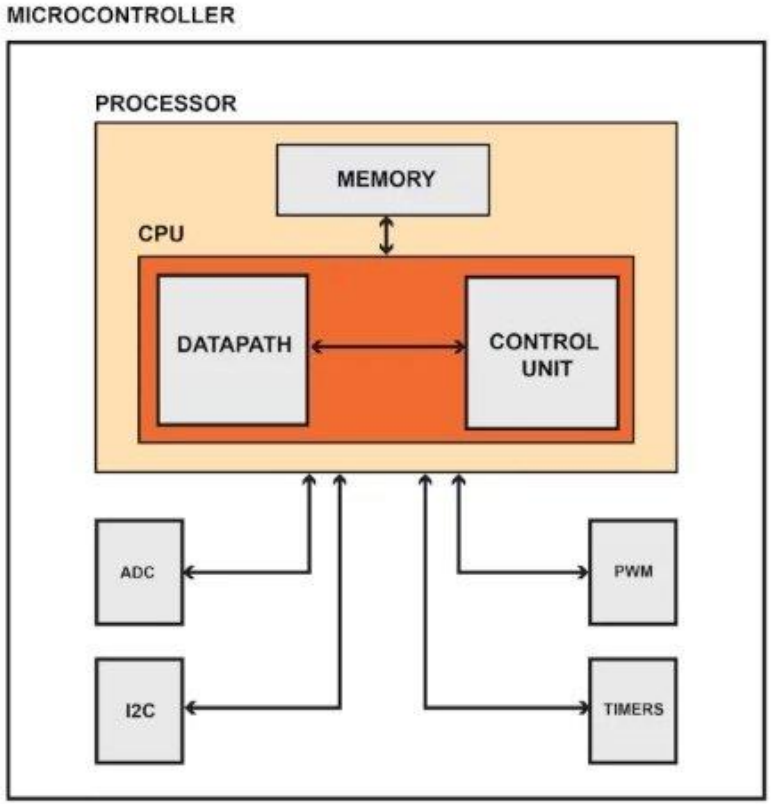
- **STM32F411- Nucleo-64**
  - Core up to 100MHz (PLL)
  - LSE: External Crystal 32kHz
  - HSI= 16MHz
  - HSE= 8MHz

- **STM32F401- Nucleo-64**
  - Core up to 84MHz (PLL)
  - LSE: External Crystal 32kHz
  - HSI= 16MHz
  - HSE= 8MHz

- **Arduino Uno/Mega**
  - Core up to 16MHz (PLL)

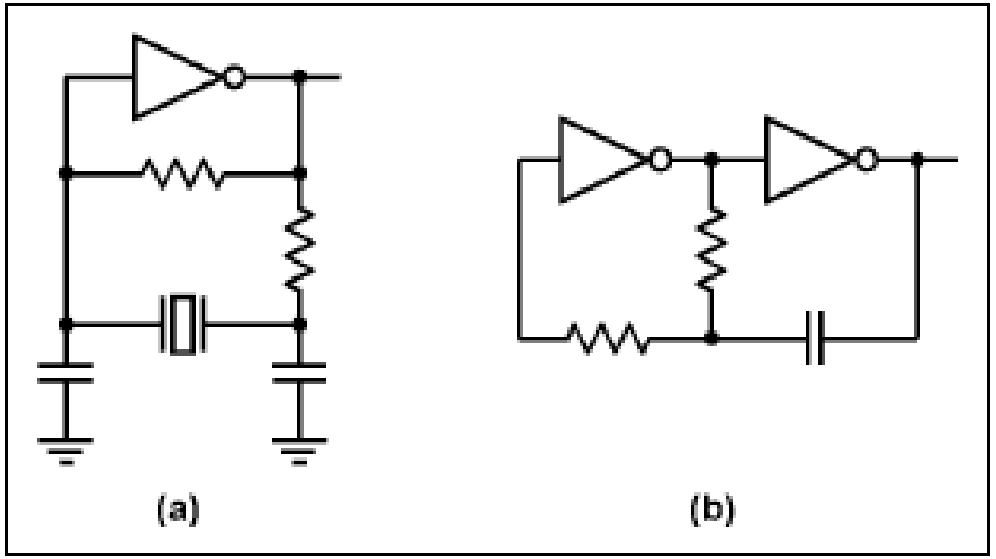
MCU System Clock

- **Peripherals need different CLK frequency for each purpose**
  - Peripheral from the same system CLK source (HSI, HSE, PLL etc)
  - Can set different clock frequency for each peripheral
    - PWM, GPIO digital Out, ADC all can have different CLK period
  - We will learn how to set the system CLK (RCC) for peripherals



HSI

- **HSI(High-Speed Internal)**
  - Generated from an internal 16 MHz RC oscillator, internal MCU
  - Can be used directly as a system clock or used as PLL input
  - Faster/Cheaper but Less Accurate than HSE
  - Must be stable: at startup, it releases CLK



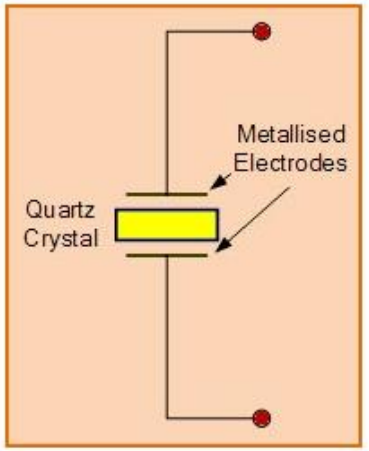
Crystal Oscillator **RC Oscillator**

HSE

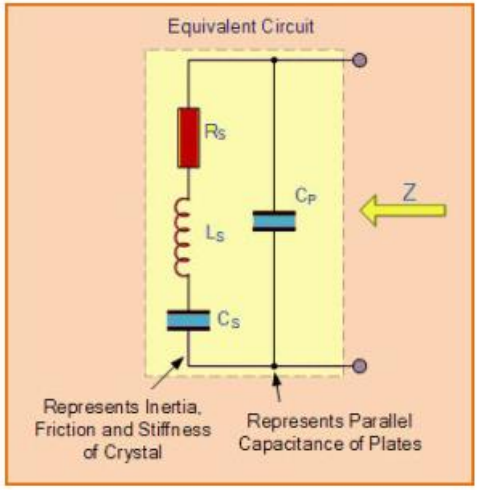
- **HSE (high-speed external)**
- The HSE has the advantage of producing a very accurate rate on the main clock
- Uses external Crystal, for stm32f411, HSE= 8MHz
- Crystal Oscillator: Quartz crystal is used in these types of oscillators
- Based on the principle of the inverse piezoelectric effect (current→oscillation)
- 8MHz crystal oscillator is commonly used in microprocessors



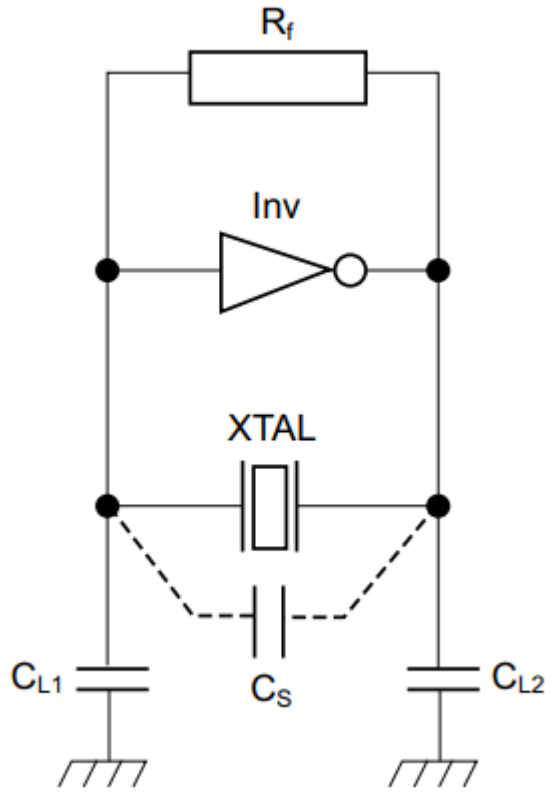
Quartz Crystal Oscillator



Electronic Symbol for Piezoelectric Crystal Resonator

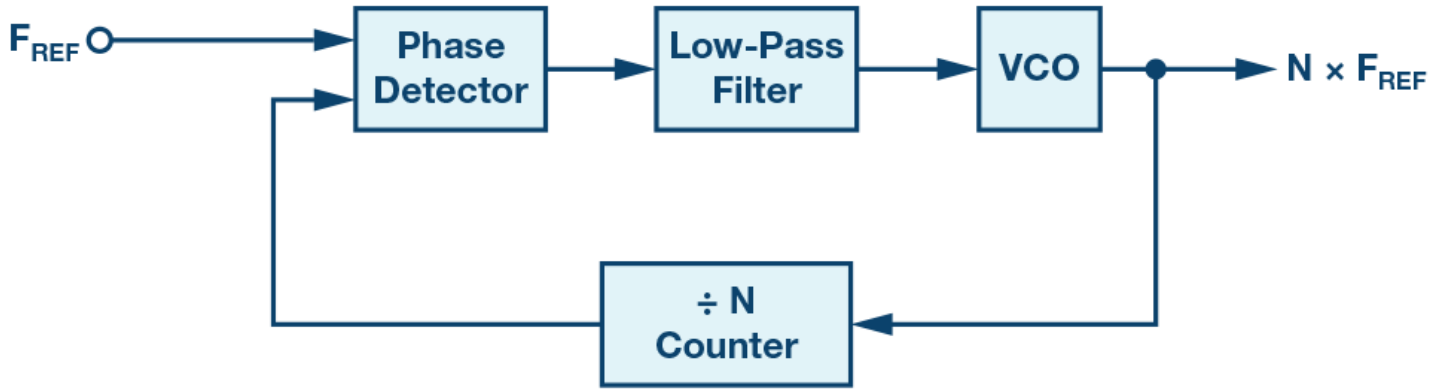


Equivalent Circuit Diagram of Quartz Crystal

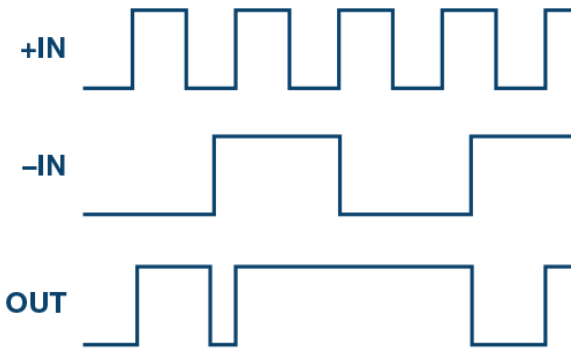


PLL

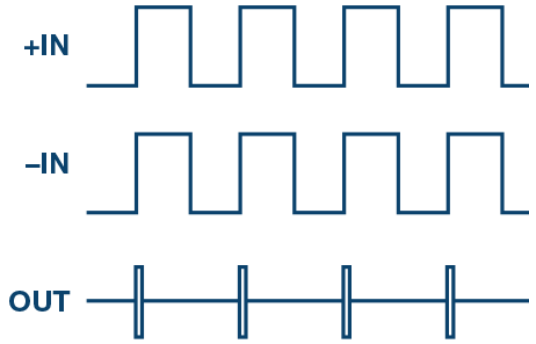
- **System Clock: PLL**
  - **Phase-locked Loop** (PLL) is a closed-loop frequency-control system based on the phase difference between the input clock signal and the feedback clock signal of a controlled oscillator (VCO)
  - phase frequency detector (PFD), charge pump,
  - loop filter, voltage-controlled-oscillator (VCO)
  - CLK source: select HSE or HSI



[analog.com]



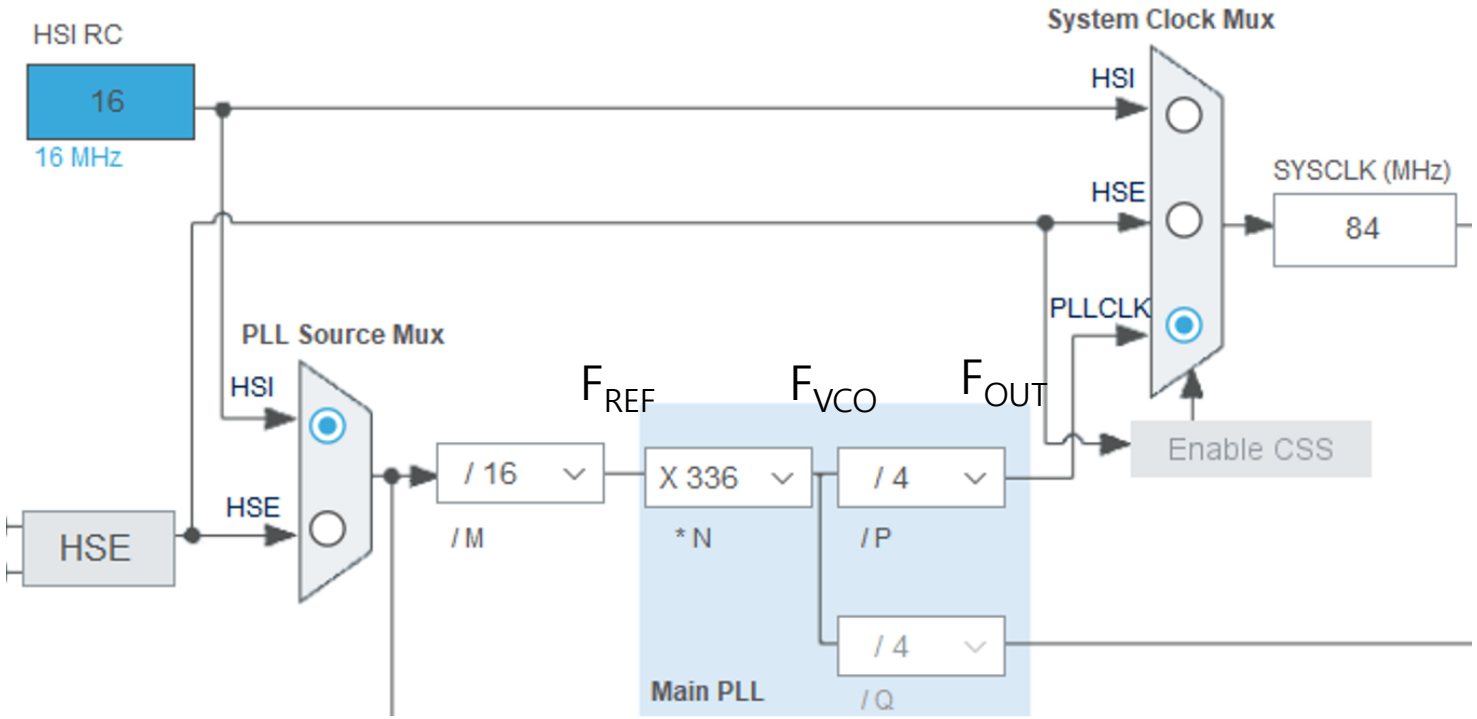
Out of phase



In phase

PLL

- **System Clock: PLL**
  - Counters: feedback counter (M), pre-scale counter (N), and post-scale counters(P)
    - $f_{(VCO\ clock)} = f_{(PLL\ clock\ input)} \times (PLLN / PLLM)$
    - $f_{(PLL\ general\ clock\ output)} = f_{(VCO\ clock)} / PLLP$



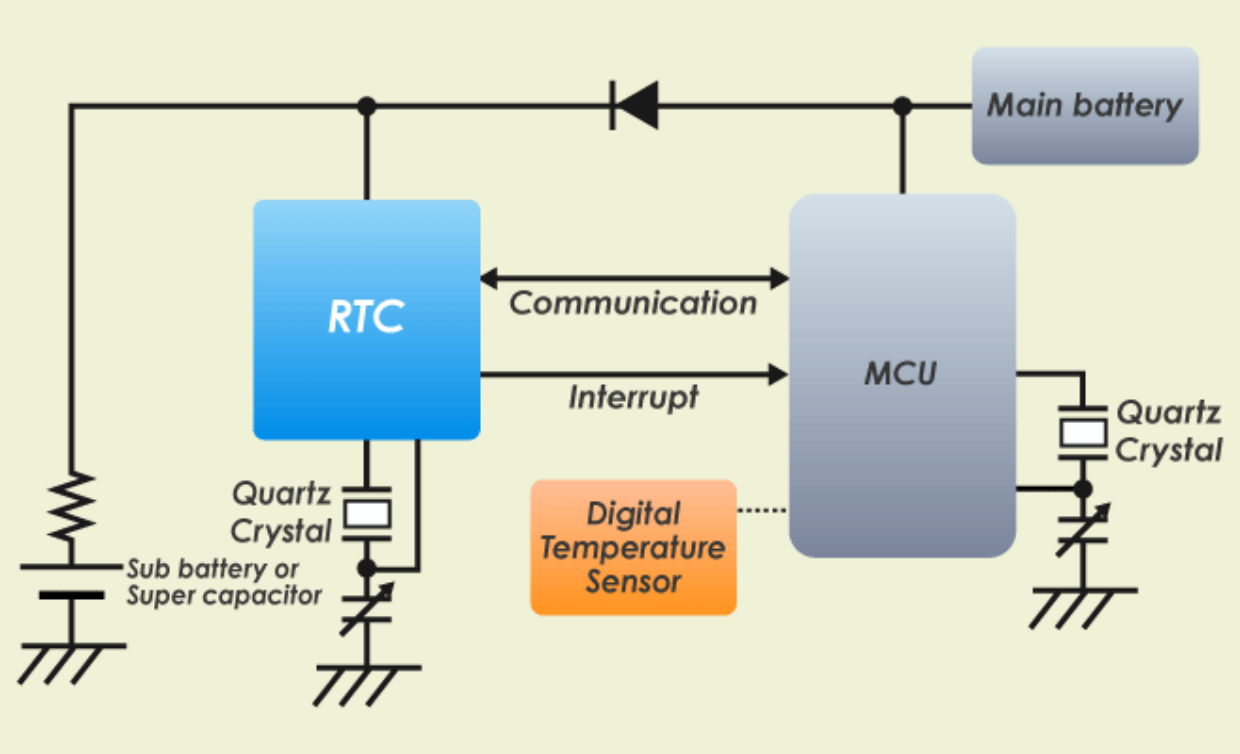
$F_{REF} = F_{IN} / M$
$F_{VCO} = F_{REF} \times N = F_{IN}(N/M)$
$F_{OUT} = F_{VCO}/P$

F_{REF} = reference
F_{VCO} = VCO frequency
F_{OUT} = output frequency

N: multiplication counter
M: divider counter
C: Post Scaler

RTC

- **Real Time Clock**
  - Real-time clock is a clock function used by systems that measure time even when the main device (such as MCU) is powered off.
    - Date/Time etc
  - 32.768kHz low-speed external crystal to drive RTC clock



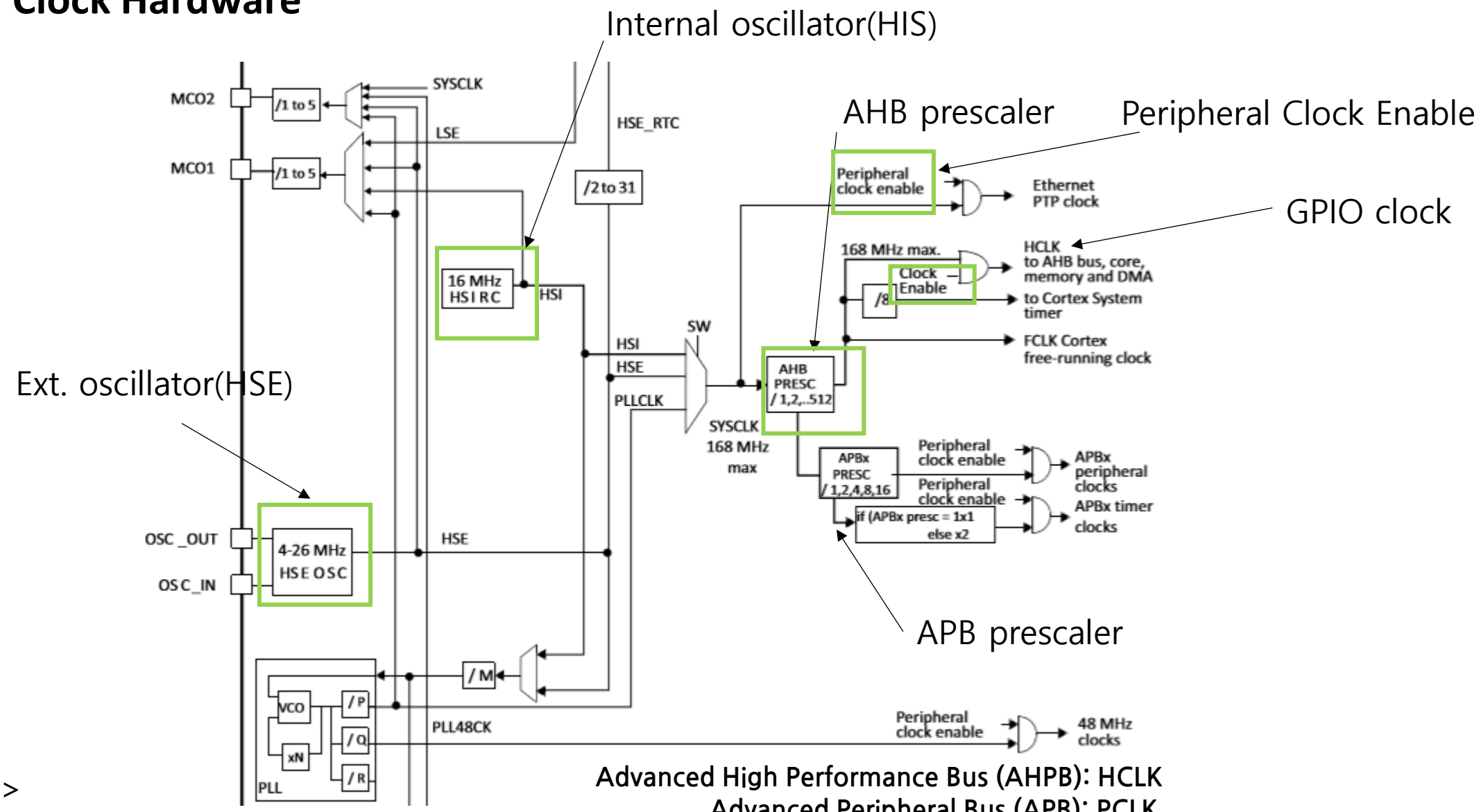
System Clock:
Reset and Clock Control (RCC)
Register Description

Embedded Controller

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RCC: Clock Control Register

- **System Clock Hardware**



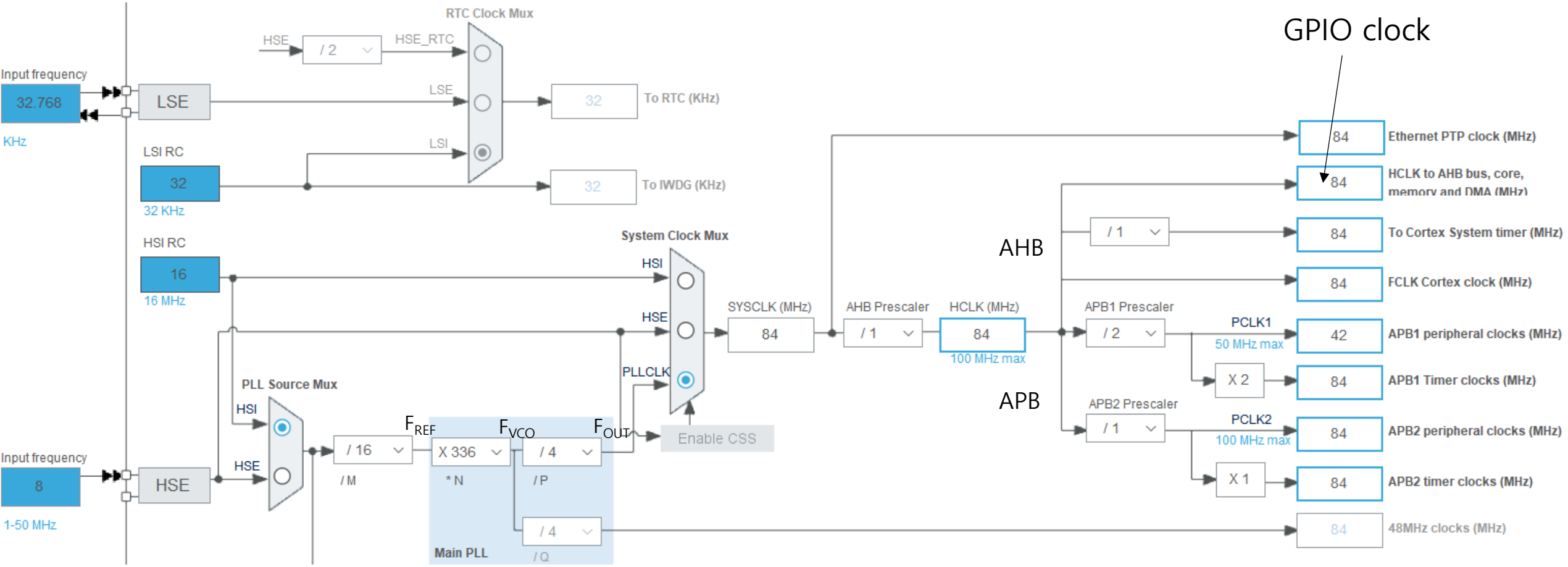
Advanced High Performance Bus (AHPB): HCLK
Advanced Peripheral Bus (APB): PCLK

<RM pg. 216>

RCC: Clock Control Register

- **Configuring PLL in CubeMx**

GPIO peripheral is AHB



RCC : How to Setup for GPIO Clock (AHB)

- **AHB (High-Speed Bus)**
  - **AHB**
    - GPIOA~H
    - DMA
    - USB

- **APB (Peripheral Bus)**
  - **APB(High speed <100MHz)**
    - TIMER
  - **APB(Low speed <50MHz)**
    - USART
    - I2C
    - SPI

RCC : How to Setup for GPIO Clock (AHB)

- **How to Setup and Enable for GPIO Clock**
  - **Option 1. HSI, Internal Clock (slower)**
    - Enable and Select HSI
    - Configure APB/AHB Prescaler
    - Enable GPIOx Clock (RCC_AHB1ENR)

  - **Option 2. PLL+HSE(HSI) (faster)**
    - Enable and Select HSE(HIS) for PLL
    - Configure APB/AHB Prescaler
    - Enable GPIOx Clock (RCC_AHB1ENR)

RCC Register

● Reset and Clock Control (RCC)

- **Clock Control**
  RCC->CR (Control Register)

- **Clock Configuration**
  RCC->CFGR (Configuration Register)
  RCC->PLLCFGR

- **Peripheral Clock Enable**
  RCC_AHB1ENR
  (AHB1 Peripheral Clock Enable Register)

<stm32f411xe.h>

```
/**
  * @brief Reset and Clock Control
  */

typedef struct
{
  __IO uint32_t CR;            /*!< RCC clock control register,
  __IO uint32_t PLLCFGR;       /*!< RCC PLL configuration register,
  __IO uint32_t CFGR;          /*!< RCC clock configuration register,
  __IO uint32_t CIR;           /*!< RCC clock interrupt register,
  __IO uint32_t AHB1RSTR;      /*!< RCC AHB1 peripheral reset register,
  __IO uint32_t AHB2RSTR;      /*!< RCC AHB2 peripheral reset register,
  __IO uint32_t AHB3RSTR;      /*!< RCC AHB3 peripheral reset register,
  uint32_t      RESERVED0;     /*!< Reserved, 0x1C
  __IO uint32_t APB1RSTR;      /*!< RCC APB1 peripheral reset register,
  __IO uint32_t APB2RSTR;      /*!< RCC APB2 peripheral reset register,
  uint32_t      RESERVED1[2];  /*!< Reserved, 0x28-0x2C
  __IO uint32_t AHB1ENR;       /*!< RCC AHB1 peripheral clock register,
  __IO uint32_t AHB2ENR;       /*!< RCC AHB2 peripheral clock register,
  __IO uint32_t AHB3ENR;       /*!< RCC AHB3 peripheral clock register,
  uint32_t      RESERVED2;     /*!< Reserved, 0x3C
  __IO uint32_t APB1ENR;       /*!< RCC APB1 peripheral clock enable register,
  __IO uint32_t APB2ENR;       /*!< RCC APB2 peripheral clock enable register,
  uint32_t      RESERVED3[2];  /*!< Reserved, 0x48-0x4C
  __IO uint32_t AHB1LPENR;     /*!< RCC AHB1 peripheral clock enable in low power mode register
  __IO uint32_t AHB2LPENR;     /*!< RCC AHB2 peripheral clock enable in low power mode register
  __IO uint32_t AHB3LPENR;     /*!< RCC AHB3 peripheral clock enable in low power mode register
  uint32_t      RESERVED4;     /*!< Reserved, 0x5C
  __IO uint32_t APB1LPENR;     /*!< RCC APB1 peripheral clock enable in low power mode register
  __IO uint32_t APB2LPENR;     /*!< RCC APB2 peripheral clock enable in low power mode register
  uint32_t      RESERVED5[2];  /*!< Reserved, 0x68-0x6C
  __IO uint32_t BDCR;          /*!< RCC Backup domain control register,
  __IO uint32_t CSR;           /*!< RCC clock control & status register,
  uint32_t      RESERVED6[2];  /*!< Reserved, 0x78-0x7C
  __IO uint32_t SSCGR;         /*!< RCC spread spectrum clock generation register,
  __IO uint32_t PLLI2SCFGR;    /*!< RCC PLLI2S configuration register,
  uint32_t      RESERVED7[1];  /*!< Reserved, 0x88
  __IO uint32_t DCKCFGR;       /*!< RCC Dedicated Clocks configuration register,
} RCC_TypeDef;
```

<Reference Manual: pg. 213>

RCC Register

- **RCC Register Memory Address**

6.3 RCC registers 161
6.3.1 RCC clock control register (RCC_CR) 161
6.3.2 RCC PLL configuration register (RCC_PLLCFGR) 163
6.3.3 RCC clock configuration register (RCC_CFGR) 165
6.3.4 RCC clock interrupt register (RCC_CIR) 167
6.3.5 RCC AHB1 peripheral reset register (RCC_AHB1RSTR) 170
6.3.6 RCC AHB2 peripheral reset register (RCC_AHB2RSTR) 173
6.3.7 RCC AHB3 peripheral reset register (RCC_AHB3RSTR) 174
6.3.8 RCC APB1 peripheral reset register (RCC_APB1RSTR) 174
6.3.9 RCC APB2 peripheral reset register (RCC_APB2RSTR) 178
6.3.10 RCC AHB1 peripheral clock register (RCC_AHB1ENR) 180
6.3.11 RCC AHB2 peripheral clock enable register (RCC_AHB2ENR) 182
6.3.12 RCC AHB3 peripheral clock enable register (RCC_AHB3ENR) 183
6.3.13 RCC APB1 peripheral clock enable register (RCC_APB1ENR) 183
6.3.14 RCC APB2 peripheral clock enable register (RCC_APB2ENR) 187
6.3.15 RCC AHB1 peripheral clock enable in low power mode register (RCC_AHB1LPENR) 189

Addr. offset	Register name	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x00	RCC_CR	Reserved		PLL SAIRDY	PLL SAION	PLL I2SRDY	PLL I2SON	PLL RDY	PLL ON	Reserved				CSSON	HSEBYP	HSERDY	HSEON	HSICAL 7	HSICAL 6	HSICAL 5	HSICAL 4	HSICAL 3	HSICAL 2	HSICAL 1	HSICAL 0	HSITRIM 4	HSITRIM 3	HSITRIM 2	HSITRIM 1	HSITRIM 0	Reserved	HSION	
0x04	RCC_PLLCFGR	Reserved				PLLQ 3	PLLQ 2	PLLQ 1	PLLQ 0	Reserved	PLLSRC	Reserved				PLLP 1	PLLP 0	Reserved	PLLN 8	PLLN 7	PLLN 6	PLLN 5	PLLN 4	PLLN 3	PLLN 2	PLLN 1	PLLN 0	PLLM 5	PLLM 4	PLLM 3	PLLM 2	PLLM 1	PLLM 0	
0x08	RCC_CFGR	MCO2 1	MCO2 0	MCO2PRE2	MCO2PRE1	MCO2PRE0	MCO1PRE2	MCO1PRE1	MCO1PRE0	I2SSRC	MCO1 1	MCO1 0	RTCPRE 4	RTCPRE 3	RTCPRE 2	RTCPRE 1	RTCPRE 0	PPRE2 2	PPRE2 1	PPRE2 0	PPRE1 2	PPRE1 1	PPRE1 0	Reserved		HPRE 3	HPRE 2	HPRE 1	HPRE 0	SWS 1	SWS 0	SW 1	SW 0	
0x0C	RCC_CIR	Reserved								CSSC	PLLSAIRDYC	PLLI2SRDYC	PLLRDYC	HSERDYC	HSIRDYC	LSERDYC	LSIRDYC	Reserved	PLLSAIRDYIE	PLLI2SRDYIE	PLLRDYIE	HSERDYIE	HSIRDYIE	LSERDYIE	LSIRDYIE	CSSF	PLLSAIRDYF	PLLI2SRDYF	PLLRDYF	HSERDYF	HSIRDYF	LSERDYF	LSIRDYF	
0x10	RCC_AHB1RSTR	Reserved		OTGHSRST	Reserved		ETHMACRST	Reserved		DMA2DRST	DMA2RST	DMA1RST	Reserved								CRCRST	Reserved	GPIOKRST	GPIOJRST	GPIOIRST	GPIOHRST	GPIOGRST	GPIOFRST	GPIOERST	GPIODRST	GPIOCRST	GPIOBRST	GPIOARST	
0x14	RCC_AHB2RSTR	Reserved																							OTGFSRS	RNGRST	HSAHRST	CRYPRST	Reserved				DCMIRST	
0x18	RCC_AHB3RSTR	Reserved																															FMCRS	
0x1C	Reserved	Reserved																																
0x20	RCC_APB1RSTR	UART8RST	UART7RST	DACRST	PWRRST	Reserved	CAN2RST	CAN1RST	Reserved	I2C3RST	I2C2RST	I2C1RST	UART5RST	UART4RST	UART3RST	UART2RST	Reserved	SPI3RST	SPI2RST	Reserved		WWDGRST	Reserved		TIM14RST	TIM13RST	TIM12RST	TIM7RST	TIM6RST	TIM5RST	TIM4RST	TIM3RST	TIM2RST	
0x24	RCC_APB2RSTR	Reserved				LTDCRST	Reserved	SAI1RST		SPI6RST	SPI5RST	Reserved	TIM11RST	TIM10RST	TIM9RST	Reserved	SYSCFGRST	SP45RST	SPI1RST	SDIORST	Reserved		ADCRST	Reserved		USART6RST	USART1RST	Reserved		TIM8RST	TIM1RST		
0x28	Reserved	Reserved																																
0x2C	Reserved	Reserved																																
0x30	RCC_AHB1ENR	Reserved	OTGHSULPIEN	OTGHSEN	ETHMACPTPEN	ETHMACRXEN	ETHMACTXEN	ETHMACEN	Reserved	DMA2DEN	DMA2EN	DMA1EN	CCMDATARAMEN	Reserved	BKPSRAMEN	Reserved					CRCEN	Reserved	GPIOKEN	GPIOJEN	GPIOIEN	GPIOHEN	GPIOGEN	GPIOFEN	GPIOEEN	GPIODEN	GPIOCEN	GPIOBEN	GPIOAEN	

RCC : Internal Clock (HSI) for GPIO

Option 1. Internal Clock (HSI) based GPIO

1. Enable HSI and choose as SYSCLK source

- Enable **HSI:** (**RCC->CR: HSION=1**)
- Wait until **HSI** is stable and ready: (**RCC->CR: HSIRDY? 1**)
- Choose the system clock switch : (**RCC->CFGR: SW = 00)**
- Check if the selected source is correct: (**RCC->CFGR: SWS ? 00)**

2. Configure HSI (optional)

- Calibration for RC oscillator: (**RCC->CR: HSICAL, HSITRIM)**

3. Configure APB/AHB Prescaler (optional)

- Change Prescaler: **RCC->CFGR: HPRE, PPRE**

4. Enable GPIOx clock(AHB1ENR)

- Enable (**RCC_AHB1ENR)** for PORTx

```
// RCC Control Register (HSI)
RCC->CR |= ((uint32_t)RCC_CR_HSION);

// wait until HSI is ready
while ( (RCC->CR & (uint32_t) RCC_CR_HSIRDY) == 0 ) {;}

// Select HSI as system clock source
// RCC Configuration Register
RCC->CFGR &= (uint32_t)((uint32_t) ~(RCC_CFGR_SW));
RCC->CFGR |= (uint32_t)RCC_CFGR_SW_HSI;

// Wait till HSI is used as system clock source
while ((RCC->CFGR & (uint32_t)RCC_CFGR_SWS) != 0 ) {;} /

// RCC Peripheral Clock Enable Register
RCC->AHB1ENR |= RCC_AHB1ENR_GPIOAEN;
```

RCC : Internal Clock (HSI) for GPIO

Option 1. Internal Clock (HSI) based GPIO

1-1. Enable HSI

- Enable **HSI** (**RCC->CR: HSION=1**)
- Wait until **HSI** is stable and ready (**RCC->CR: HSIRDY? 1**)

Bit 1 **HSIRDY**: Internal high-speed clock ready flag

0: HSI oscillator not ready
1: HSI oscillator ready

Bit 0 **HSION**: Internal high-speed clock enable

0: HSI oscillator OFF
1: HSI oscillator ON

6.3.1 RCC clock control register (RCC_CR)

Address offset: 0x00

Reset value: 0x0000 XX81 where X is undefined.

Access: no wait state, word, half-word and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				PLLI2S RDY	PLLI2S ON	PLLRDY	PLLON	Reserved				CSS ON	HSE BYP	HSE RDY	HSE ON
				r	rw	r	rw					rw	rw	r	rw

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HSICAL[7:0]								HSITRIM[4:0]					Res.	HSI RDY	HSION
r	r	r	r	r	r	r	r	rw	rw	rw	rw	rw		r	rw

read

read/write

RCC : Internal Clock (HSI) for GPIO

Option 1. Internal Clock (HSI) based GPIO

1-2. Choose HSI as SYSCLK source:

- Choose the system clock switch : (**RCC->CFGR : SW = 00**)
- Check if the selected source is correct: (**RCC->CFGR : SWS ? 00**)

```
RCC->CFGR &= (uint32_t)(~RCC_CFGR_SW);
RCC->CFGR |= (uint32_t)RCC_CFGR_SW_HSI;

while ((RCC->CFGR & (uint32_t)RCC_CFGR_SWS) != 0 );
```

6.3.3 RCC clock configuration register (RCC_CFGR)

Address offset: 0x08

Reset value: 0x0000 0000

Access: 0 ≤ wait state ≤ 2, word, half-word and byte access

1 or 2 wait states inserted only if the access occurs during a clock source switch.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MCO2		MCO2 PRE[2:0]			MCO1 PRE[2:0]			I2SSCR	MCO1		RTCPRE[4:0]				
rw		rw	rw	rw	rw	rw	rw	rw	rw		rw	rw	rw	rw	rw

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PPRE2[2:0]			PPRE1[2:0]			Reserved		HPRE[3:0]				SWS1	SWS0	SW1	SW0
rw	rw	rw	rw	rw	rw			rw	rw	rw	rw	r	r	rw	rw

Bits 3:2 **SWS:** System clock switch status

Set and cleared by hardware to indicate which clock source is used as the system clock.

00: HSI oscillator used as the system clock

01: HSE oscillator used as the system clock

10: PLL used as the system clock

11: not applicable

Bits 1:0 **SW:** System clock switch

Set and cleared by software to select the system clock source.

Set by hardware to force the HSI selection when leaving the Stop or Standby mode or in case of failure of the HSE oscillator used directly or indirectly as the system clock.

00: HSI oscillator selected as system clock

01: HSE oscillator selected as system clock

10: PLL selected as system clock

11: not allowed

RCC : Internal Clock (HSI) for GPIO

Option 1. Internal Clock (HSI) based GPIO

3. (optional) Configure APB/AHB Prescaler

Change Prescaler: **(RCC->CFGR : HPRE, PPRE)**

Bits 7:4 **HPRE:** AHB prescaler

Set and cleared by software to control AHB clock division factor.

Caution: The clocks are divided with the new prescaler factor from 1 to 16 AHB cycles after HPRE write.

0xxx: system clock not divided
1000: system clock divided by 2
1001: system clock divided by 4
1010: system clock divided by 8
1011: system clock divided by 16
1100: system clock divided by 64
1101: system clock divided by 128
1110: system clock divided by 256
1111: system clock divided by 512

AHB clock = system clock = 84MHz

```
RCC->CFGR &= ~RCC_CFGR_HPRE;
RCC->CFGR &= ~RCC_CFGR_PPRE1;
RCC->CFGR |=  RCC_CFGR_PPRE1_2;
RCC->CFGR &= ~RCC_CFGR_PPRE2;
```

Bits 15:13 **PPRE2:** APB high-speed prescaler (APB2)

Set and cleared by software to control APB high-speed clock division factor.

Caution: The software has to set these bits correctly not to exceed 100 MHz on this domain. The clocks are divided with the new prescaler factor from 1 to 16 AHB cycles after PPRE2 write.

0xx: AHB clock not divided
100: AHB clock divided by 2
101: AHB clock divided by 4
110: AHB clock divided by 8
111: AHB clock divided by 16

APB2 clock = 84MHz/1 < 100MHz

Bits 12:10 **PPRE1:** APB Low speed prescaler (APB1)

Set and cleared by software to control APB low-speed clock division factor.

Caution: The software has to set these bits correctly not to exceed 50 MHz on this domain. The clocks are divided with the new prescaler factor from 1 to 16 AHB cycles after PPRE1 write.

0xx: AHB clock not divided
100: AHB clock divided by 2
101: AHB clock divided by 4
110: AHB clock divided by 8
111: AHB clock divided by 16

APB1 clock = 84MHz / 2 = 42MHz < 50MHz

RCC : Internal Clock (HSI) for GPIO

Option 1. Internal Clock (HSI) based GPIO

4. Enable GPIOx clock

- Enable (**RCC_AHB1ENR)** for PORTx

6.3.9 RCC AHB1 peripheral clock enable register (RCC_AHB1ENR)

Address offset: 0x30

Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access.

```
RCC->AHB1ENR |= RCC_AHB1ENR_GPIOAEN;
```

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved									DMA2EN	DMA1EN	Reserved				
									rw	rw					

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			CRCEN	Reserved				GPIOHEN	Reserved		GPIOEEN	GPIODEN	GPIOCEN	GPIOBEN	GPIOAEN
			rw					rw			rw	rw	rw	rw	rw

Bit 0 **GPIOAEN:** IO port A clock enable
Set and cleared by software.
0: IO port A clock disabled
1: IO port A clock enabled

RCC : PLL for GPIO

Option 2. PLL based GPIO Clock

1. Enable either (HSI, or HSE) for PLL and Choose PLL for System Clock

- Enable **HSE or HSI** — **RCC->CR : HSION=1**
- Wait until **HSE** or **HSI** is stable — **RCC->CR : HSIRDY? 1**
- Choose PLL for system clock switch : — **RCC->CFGR : SW = 10**
- Check if PLL selection is correct: — **RCC->CFGR : SWS ? 10**

2. Select the clock source for PLL

- Select the PLL source(HSI or HSE): — **RCC->PLLCFGR : PLLSRC= 0 or 1**

3. (Optional)Configure PLL parameters

- Select (M/N/P): — **RCC->PLLCFGR : PLLM, PLLN, …**

4. Enable PLL

- Enable main PLL: — **RCC->CR : PLLON=1**

5. (Optional)Configure APB/AHB Prescaler

- Change Prescaler: — **RCC->CFGR : HPRE, PPRE**

6. Enable GPIOx clock(AHB1ENR)

- Enable (**RCC_AHB1ENR)** for PORTx

RCC : PLL for GPIO

Option 2. PLL based GPIO Clock

1. Enable either (HSI, or HSE) for PLL and Choose PLL for System Clock

- Enable **HSE or HSI** — **RCC->CR : HSION=1**
- Wait until **HSE** or **HSI** is stable — **RCC->CR : HSIRDY? 1**
- Choose PLL for system clock switch : — **RCC->CFGR : SW = 10**
- Check if PLL selection is correct: — **RCC->CFGR : SWS ? 10**

6.3.3 RCC clock configuration register (RCC_CFGR)

Address offset: 0x08

Reset value: 0x0000 0000

Access: 0 ≤ wait state ≤ 2, word, half-word and byte access

1 or 2 wait states inserted only if the access occurs during a clock source switch.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MCO2		MCO2 PRE[2:0]			MCO1 PRE[2:0]			I2SSCR	MCO1		RTCPRE[4:0]				
rw		rw	rw	rw	rw	rw	rw	rw	rw		rw	rw	rw	rw	rw

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PPRE2[2:0]			PPRE1[2:0]			Reserved		HPRE[3:0]				SWS1	SWS0	SW1	SW0
rw	rw	rw	rw	rw	rw			rw	rw	rw	rw	r	r	rw	rw

Bits 3:2 **SWS:** System clock switch status

Set and cleared by hardware to indicate which clock source is used as the system clock.

00: HSI oscillator used as the system clock

01: HSE oscillator used as the system clock

10: PLL used as the system clock

11: not applicable

Bits 1:0 **SW:** System clock switch

Set and cleared by software to select the system clock source.

Set by hardware to force the HSI selection when leaving the Stop or Standby mode or in case of failure of the HSE oscillator used directly or indirectly as the system clock.

00: HSI oscillator selected as system clock

01: HSE oscillator selected as system clock

10: PLL selected as system clock

11: not allowed

RCC : PLL for GPIO

Option 2. PLL based GPIO Clock

2. Select the clock source for PLL

- Select the PLL source(HSI or HSE): **RCC->PLLCFGR : PLLSRC= 0 or 1**
- Need to disable PLL before selection

6.3.2 RCC PLL configuration register (RCC_PLLCFGR)

Address offset: 0x04

Reset value: 0x2400 3010

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				PLLQ3	PLLQ2	PLLQ1	PLLQ0	Reserved	PLLSRC	Reserved				PLLP1	PLLP0
				rw	rw	rw	rw		rw					rw	rw

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	PLLN									PLLM5	PLLM4	PLLM3	PLLM2	PLLM1	PLLM0
	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 22 **PLLSRC:** Main PLL(PLL) and audio PLL (PLLI2S) entry clock source

Set and cleared by software to select PLL and PLLI2S clock source. This bit can be written only when PLL and PLLI2S are disabled.

0: HSI clock selected as PLL and PLLI2S clock entry

1: HSE oscillator clock selected as PLL and PLLI2S clock entry

Example:

(PLL clock input source) = HSI clock = 16 MHz

```
RCC->PLLCFGR &= ~RCC_PLLCFGR_PLLSRC;
RCC->PLLCFGR |= RCC_PLLCFGR_PLLSRC_HSI;
```

RCC : PLL for GPIO

Option 2. PLL based GPIO Clock

3. (Optional)Configure PLL parameters

***p164 [RM]**

- Select (M/N/P): **RCC->PLLCFGR: PLLM, PLLN,** …
- Caution: VCO output frequency is between 100 and 432 MHz. (50 ≤ PLLN ≤ 432)
- Caution: VCO input frequency ranges from 1 to 2 MHz(recommend). (PLLM with 2 ≤ PLLM ≤ 63)
- Caution: PLL do not exceed 80MHz for this MCU (PLLP=2,4,6,8)

Bits 14:6 **PLLN:** Main PLL (PLL) multiplication factor for VCO

Bits 5:0 **PLLM:** Division factor for the main PLL (PLL) and audio PLL (PLLI2S) input clock

Bits 17:16 **PLLP:** Main PLL (PLL) division factor for main system clock

000110010: PLLN = 50 to 110110000: PLLN = 432

000010: PLLM = 2 to 111111: PLLM = 63

6.3.2 RCC PLL configuration register (RCC_PLLCFGR)

This register is used to configure the PLL clock outputs according to the formulas:

- $f_{(VCO\ clock)} = f_{(PLL\ clock\ input)} \times (PLLN / PLLM)$
- $f_{(PLL\ general\ clock\ output)} = f_{(VCO\ clock)} / PLLP$
- $f_{(USB\ OTG\ FS,\ SDIO)} = f_{(VCO\ clock)} / PLLQ$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				PLLQ3	PLLQ2	PLLQ1	PLLQ0	Reserved	PLLSRC	Reserved				PLLP1	PLLP0
				rw	rw	rw	rw		rw					rw	rw

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	PLLN									PLLM5	PLLM4	PLLM3	PLLM2	PLLM1	PLLM0
	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Example:
PLL clock source(HSI) = 16 MHz
PLLN=50
PLLM=8
PLLP=2

→ PLL clock=[(16/8)*50]/2=50MHz

RCC : PLL for GPIO

Option 2. PLL based GPIO Clock

3. (Optional)Configure PLL parameters

Bits 5:0 **PLLM:** Division factor for the main PLL (PLL) input clock

Set and cleared by software to divide the PLL input clock before the VCO. These bits can be written only when the PLL is disabled.

Caution: The software has to set these bits correctly to ensure that the VCO input frequency ranges from 1 to 2 MHz. It is recommended to select a frequency of 2 MHz to limit PLL jitter.

VCO input frequency = PLL input clock frequency / PLLM with 2 ≤PLLM ≤63

000000: PLLM = 0, wrong configuration
000001: PLLM = 1, wrong configuration
000010: PLLM = 2
000011: PLLM = 3
000100: PLLM = 4

PLLM = 8

Bits 14:6 **PLLN:** Main PLL (PLL) multiplication factor for VCO

Set and cleared by software to control the multiplication factor of the VCO. These bits can be written only when PLL is disabled. Only half-word and word accesses are allowed to write these bits.

Caution: The software has to set these bits correctly to ensure that the VCO output frequency is between 100 and 432 MHz. (check also *Section 6.3.20: RCC PLLI2S configuration register (RCC_PLLI2SCFGR)*)

VCO output frequency = VCO input frequency × PLLN with 50 ≤PLLN ≤432

000000000: PLLN = 0, wrong configuration
000000001: PLLN = 1, wrong configuration
...
000110010: PLLN = 50
...
001100011: PLLN = 99
001100100: PLLN = 100
...

PLLN = 84

Bits 17:16 **PLLP:** Main PLL (PLL) division factor for main system clock

Set and cleared by software to control the frequency of the general PLL output clock. These bits can be written only if PLL is disabled.

Caution: The software has to set these bits correctly not to exceed 100 MHz on this domain.

PLL output clock frequency = VCO frequency / PLLP with PLLP = 2, 4, 6, or 8

00: PLLP = 2
01: PLLP = 4
10: PLLP = 6
11: PLLP = 8

- f(VCO clock) = f(PLL clock input) x (PLLN/PLLM)
  = 16MHz x (84 / 8)
  = 168MHz
- f(PLL general clock output) = f(VCO clock) / PLLP
  = 168MHz / 2
  = **84MHz**

```
RCC->PLLCFGR = (RCC->PLLCFGR & ~RCC_PLLCFGR_PLLN) | 84U << 6;
RCC->PLLCFGR = (RCC->PLLCFGR & ~RCC_PLLCFGR_PLLM) | 8U ;
RCC->PLLCFGR &= ~RCC_PLLCFGR_PLLP;
```

RCC : PLL for GPIO

Option 2. PLL based GPIO Clock

4. Enable PLL

- Enable main PLL: **RCC->CR: PLLON=1**

Bit 25 **PLLRDY**: Main PLL (PLL) clock ready flag
Set by hardware to indicate that PLL is locked.
0: PLL unlocked
1: PLL locked

Bit 24 **PLLON**: Main PLL (PLL) enable
Set and cleared by software to enable PLL.
Cleared by hardware when entering Stop or Standby mode. This bit cannot be reset if PL
clock is used as the system clock.
0: PLL OFF
1: PLL ON

6.3.1 RCC clock control register (RCC_CR)

Address offset: 0x00

Reset value: 0x0000 XX81 where X is undefined.

Access: no wait state, word, half-word and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				PLLI2S RDY	PLLI2S ON	PLLRDY	PLLON	Reserved				CSS ON	HSE BYP	HSE RDY	HSE ON
				r	rw	r	rw					rw	rw	r	rw

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HSICAL[7:0]								HSITRIM[4:0]					Res.	HSI RDY	HSION
r	r	r	r	r	r	r	r	rw	rw	rw	rw	rw		r	rw